

Fig. 5.4 Graphs of trajectories obtained by choosing three different initial conditions on the energy surface $H = 4$. The initial conditions are as follows: (*left*) $\boldsymbol{q}_0 = (1.75, 0); \boldsymbol{p}_0 = (0, 1.1748)$, (*center*) $\boldsymbol{q}_0 = (0, -2.9); \boldsymbol{p}_0 = (0.8718, 0)$, (*right*) $\boldsymbol{q}_0 = (0, -2.7), \boldsymbol{p}_0 = (0, 1.4283)$. The numerical method (the Verlet method) used 100 million steps with stepsize $\Delta t = 0.0001$

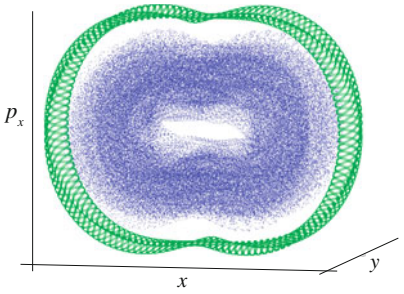


Fig. 5.5 Projecting the data from two of the initial conditions on the $H = 4$ energy surface onto the x, y, p_x subspace reveals the accessible set to be fragmented by 2-tori; the system clearly fails to be ergodic on the energy surface

projection of one of the orbits in the x, y, p_x coordinates and is shown in Fig. 5.5.

Example 5.6 (Anisotropic Oscillator) The anisotropic oscillator was used in Chap. 1 to illustrate the concept of a chaotic dynamical system. Fix, arbitrarily, a small set of initial conditions for the anisotropic pendulum at energy E_0, with positions in a small interval on the positive x axis, $1.05 \leq x \leq 1.15$ and $y = 0$. For each point $(x, 0)$ taken from this set select the initial velocities as $\dot{x} = 0$, $\dot{y} = [2(\mathrm{E}_0 - U(x, 0))]^{1/2}$. The set $y = 0, \dot{y} = 0$ is invariant under the differential equations of the perturbed central force problem, since the force acts in the radial direction; our choice of initial conditions ensures that trajectories explore the larger $H = \mathrm{E}_0$ energy surface.

The set of initial conditions forms a line segment in the four-dimensional phase space. These points are then integrated forward in time by solving the differential equations (using a numerical method), and the resulting image sets are shown in Fig. 5.6. The first panel of Fig. 5.6 shows, in one set of axes, the image of the initial set at times $t = 4, 8, 12, 16$. Successive figures show the images of the initial set after times $t = 20, 28, 36$. The line segment becomes curved, and spreads out,